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Clinical Genomic Report

Patient ID: PAT001-OVC-2025 **Age/Sex:** 58F

Cancer Type: Ovarian Cancer (High-Grade Serous Carcinoma (HGSOC)) **Stage:** Stage IV

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1. Genomic Profile

#	GENE	VARIANT	VAF	SIGNIFICANCE	ACTIONABILITY
1	**BRCA1**	Pathogenic germline mutation (VAF 0.48)	48%	Pathogenic	Actionable FDA-approved PARP inhibitors: olaparib (Lynparza), niraparib (Zejula)
2	**TP53**	R175H missense (VAF 0.73)	73%	Pathogenic	Actionable Neoantigen peptide vaccine trials (investigational)
3	**PIK3CA**	E545K missense (VAF 0.42)	42%	Pathogenic	Actionable Alpelisib — ovarian off-label trials active
4	**CCNE1**	Amplification (19q12, log2=1.12)	—	Pathogenic	Actionable Adavosertib (Wee1 inhibitor) — Phase II trials in CCNE1-amplified HGSOC
5	**MYC**	Amplification (8q24, log2=1.85)	—	Pathogenic	Actionable BET inhibitors (OTX015) — Phase I

3. Spatial Transcriptomics

900 spots analyzed. Tumor 20%, stromal 51%, immune 18%. CD8:Treg ratio=1.17. CD8A Moran's I=0.061.

Immune Status: **MIXED** — Heterogeneous infiltration

- CD8A weakly clustered (Moran's I=0.061) — immune-excluded phenotype
- Stromal compartment 51% — large barrier to immune infiltration
- CD8:Treg ratio 1.17 (borderline)

4. GEARS Perturbation Predictions

Treatment: NNMT+STAT3 knockdown | **Cell type:** tumor_core

5. Treatment Hypotheses

1 Olaparib (Lynparza) or Niraparib (Zejula) FDA Approved

Type: targeted | **Response:** 60-70% in BRCA-mutant HGSOC

Rationale: BRCA1 germline pathogenic mutation + HRD score 72 provides dual eligibility.

2 Alpelisib (Piqray) ESCAT III

Type: targeted | **Response:** 35-40% in PIK3CA-mutant tumors

Rationale: PIK3CA E545K + PTEN loss creates compounded PI3K pathway activation.

Type: targeted | **Response:** 30-40% in CCNE1-amplified HGSOC

Rationale: CCNE1 amplification (log2=1.12) is the target biomarker.

4 **Neoantigen Peptide Vaccine (RMPEAAPPV)** Clinical Trial Only

Type: immunotherapy

Rationale: RMPEAAPPV confirmed strong HLA-A*02:01 binder (IC50=7.8 nM, top 0.07 percentile).

5 **NNMT/STAT3 Disruption + Checkpoint Inhibitor** Clinical Trial Only

Type: immunotherapy

Rationale: Spatial deconvolution shows large stromal compartment (34%) consistent with immune-excluded phenotype.

6. Open Targets Drug-Target Associations

#	GENE	OVERALL SCORE	DISEASE	EVIDENCE	TRACTABILITY	KNOWN DRUGS
1	**PIK3CA**	0.82	—	—	—	—
2	**CCNE1**	0.28	—	—	—	—
3	**MYC**	0.24	—	—	—	—

7. XAI Evidence Strength Summary

HIGH 0 MODERATE 7 LOW 1

ANALYSIS COMPONENT	CONFIDENCE	EVIDENCE GRADE	NOTE
Cell-Type Deconvolution	MODERATE	Algorithm-Predicted — Not Clinical Grade	Signature-based deconvolution across 900 spots using HGSOC marker panels.
Copy Number Variants	MODERATE	Algorithm-Predicted — Not Clinical Grade	CNVkit panel-based segmentation. Limited resolution for arm-level events.
HR Deficiency Score	MODERATE	Algorithm-Predicted — Not Clinical Grade	Algorithmic HRD estimate (LOH+TAI+LST proxy). Not Myriad myChoice CDx validated.
MHC I Neoantigen Binding	MODERATE	Computational Prediction — Research Only	RMPEAAPPV IC50=7.8 nM (strong, top 0.07 percentile). NetMHCpan accuracy ~70-85%.
Neoantigen Burden	MODERATE	Computational Prediction — Research Only	TMB-to-neoantigen conversion (4.2 mut/Mb x HGSOC factor 12 = ~50 candidates).
Drug Perturbation Response	LOW	Computational Prediction — Research Only	GEARS GNN dry-run: canonical synthetic payload, not patient-specific inference.
Somatic Variant Calls	MODERATE	Computational Prediction — Research Only	Variants reconstructed from report table; synthetic fallback used. Not from raw VCF/FASTQ.
Spatial Autocorrelation (Moran's I)	MODERATE	Algorithm-Predicted — Not Clinical Grade	Moran's I on 900 spots. Significant but weak effect sizes.

Synthetic Data Warning: The following analyses used synthetic or simulated data and carry **no clinical validity**:

- Somatic Variant Calls
- Drug Perturbation Response

Action Required: This report contains 1 low-confidence item requiring clinical review:

- Drug Perturbation Response

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be reviewed by a qualified oncologist before any treatment decisions.

Data Sources: mcp-genomic-results, mcp-neoantigen, mcp-spatialtools, mcp-opentargets, mcp-perturbation, mcp-patient-report | **Report Version:** 3.1 | **Generated:** 2026-07-10 09:22:38